

Visualizing Lattice-based Post-Quantum Cryptographic Algorithms

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Abstract. Post-quantum cryptographic algorithms are gaining relevance as The abstract should briefly summarize the contents of the paper in 150–250 words.

Keywords: Cryptography · Post-Quantum · Another keyword.

1 Introduction

Cryptography is the study of techniques and protocols relating to securing communications (Rivest, 1990). Typically, this consists of constructing and analyzing protocols that prevent outside parties from reading private messages.

Prior to the modern era, most of cryptography was encryption, which is a process that transforms some form of information (plaintext) into an alternative form that only certain parties can decode (ciphertext). What is now known as classical cryptography consisted of using ciphers that could be practically computed and solved by hand. However, after devices such as the Enigma machine and other computational devices were developed, these ciphers became trivially easy to break.

Now, history repeats itself with the development of quantum computers as they threaten to render multiple modern cryptographic algorithms useless. Quantum algorithms such as the ones proposed by Peter Shor are theorized to be able to break protocols like RSA given a strong enough quantum computer [3]. Because of this, cryptology experts around the world have looked into new protocols that might have the potential to defend against them.

NIST (National Institute of Standards and Technology) is one of the organizations at the forefront of the effort. In 2016, they put out a call for proposals that might be used in post-quantum cryptography. Eight years later, they released three standards based on those submissions, two of which are lattice-based; ML-KEM and ML-DSA, based on the Kyber and Dilithium algorithms respectively. ML-KEM is used for key encapsulation, while ML-DSA is used for digital signatures.

Despite their importance, there are a limited number of resources available for learning about them. In some part this is because they are relatively new standards. However, it can also be attributed to their complexity and nature as

lattice-based. Aside from NIST’s released papers (NIST, 2024), attempts have been made to visualize the scheme and other lattice-based algorithms, including the following:

- pqc-vizz (ML-KEM, ML-DSA, SLH-DSA)
- CrypTool 2 (cryptography module)

However, they tend to skip parts of the step-by-step process, not show the background calculations, as well as simplify the lattice too much (Rodríguez et al., 2024; Kalava et al., 2025).

Thus, this work aims to develop a visualization tool that allows users to view the complete process behind ML-KEM and ML-DSA.

2 Related Works

TODO: The idea of using visualization tools to facilitate the understanding of complex topics is not new. Visualization tools for topics such as THIS have been . However, the current tools available for visualizing ML-KEM and ML-DSA (which are ultimately lattice-based algorithms) are limited.

2.1 ML-KEM

ML-KEM, also referred to as Module-Lattice-Based Key-Encapsulation Mechanism Standard, is a standard set by NIST’s FIPS 203 publication that specifies a set of algorithms meant to be used to establish a shared secret key between two parties. This standard was proposed as a response to the potential unreliability of modern key generation algorithms such as RSA and AES in the face of quantum computers.

ML-KEM is a variant of the CRYSTALS-Kyber algorithm, which is itself based on the hardness of solving the module learning with errors (MLWE) problem.

In brief, Kyber follows a two stage approach: employing an IND-CPA secure public key encryption scheme that encrypts messages of a fixed length of 32 bytes, followed by a slightly tweaked Fujisaki Okamoto (FO) transform to construct the IND-CCA2 KEM.

2.2 ML-DSA

Explain ML-KEM and how it works ML-DSA, also known as Module-Lattice-Based Digital Signature Standard, is a standard set by NIST’s FIPS 204 publication that specifies a set of algorithms for

2.3 Existing Visualizations

There are a number of existing visualization tools for ML-KEM, ML-DSA, and lattices in general. They have a tendency to fall into one of the following categories.

Examples of existing visualizations and pros and cons. Categories probably? The ones who use dots, the ones who use colors, the ones who use arrows, the ones who use matrices, etcetc

2.3.1 Dot Visualization

Dot visualizations represent lattices as points in space, either on

2.3.2 Color Visualization

Lattices can be visualized through the usage of different colors, each representing a different value. The pqc-vizz tool uses colored grids in addition to matrices to represent lattices. Each value is assigned a corresponding hue, saturation, and lightness through the following formula:

```
const getColor = (value: number) => {
  const hue = (value / 255) * 240;
  const saturation = 80;
  const lightness = 40 + (value / 255) * 40;
  return `hsl(${hue}, ${saturation}%, ${lightness}%)`;
};
```

Below is an example of a lattice visualized in that way, along with the values that each color represents.

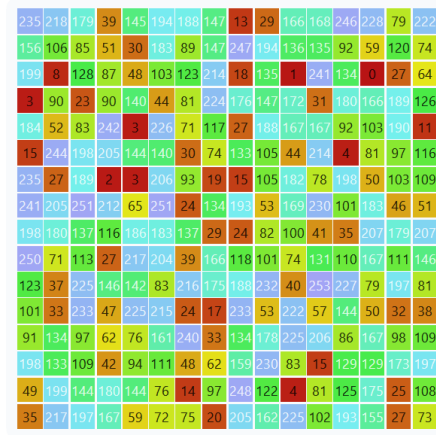


Fig. 1. Representation of a lattice through colors

2.3.3 Matrix Visualization

3 Visualization Tool Design

The developed visualization tool is a web application that provides visualizations for ML-KEM, ML-DSA, LLL, and BKZ. (elaborate later)

3.1 Architecture

The visualization tool is built using Next.js, a React-based framework, for the frontend, and primarily Typescript for the backend processes. The ML-KEM and ML-DSA algorithms are implemented using an existing Javascript library and imported as needed. (insert diagram later + be more thorough)

3.2 ML-KEM

The implementation of ML-KEM comes from

3.3 ML-DSA

This part is for explaining how ML-DSA is implemented/how the code works

4 Results?

5 Conclusion

Theorem 1. *This is a sample theorem. The run-in heading is set in bold, while the following text appears in italics. Definitions, lemmas, propositions, and corollaries are styled the same way.*

Proof. Proofs, examples, and remarks have the initial word in italics, while the following text appears in normal font.

For citations of references, we prefer the use of square brackets and consecutive numbers. Citations using labels or the author/year convention are also acceptable. The following bibliography provides a sample reference list with entries for journal articles [?], an LNCS chapter [?], a book [?], proceedings without editors [?], and a homepage [?]. Multiple citations are grouped [?,?,?], [?,?,?,?].

5.0.1 Acknowledgments. A bold run-in heading in small font size at the end of the paper is used for general acknowledgments, for example: This study was funded by X (grant number Y).

5.0.2 Disclosure of Interests. It is now necessary to declare any competing interests or to specifically state that the authors have no competing interests. Please place the statement with a bold run-in heading in small font size beneath the (optional) acknowledgments¹, for example: The authors have no competing interests to declare that are relevant to the content of this article. Or: Author A has received research grants from Company W. Author B has received a speaker honorarium from Company X and owns stock in Company Y. Author C is a member of committee Z.

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